



# North American Tundra

*Stretching across the continent's northern reaches, a cold, treeless landscape where plants huddle to the ground to survive*



## △ POLYGONAL PATTERN

The continual Arctic freeze-thaw cycle creates ice wedges that eventually shape tundra soil into polygons. These become freshwater ponds during the summer thaw.

In North America, tundra covers northern Alaska in the U.S., extends eastward through northern Canada, and forms a narrow band around the coast of Greenland. North of the Arctic Circle, this ecoregion is known as Arctic tundra, while in the mountains above the timberline, it is classified as alpine tundra.

## Challenging conditions

Arctic tundra conditions are the coldest: summer high temperatures are just 40°F (4°C), plunging to -25°F (-32°C) or below in winter. The alpine tundra climate is less extreme: summers may reach 54°F (12°C) while winters rarely fall below 0°F (-18°C). However, in both cases, this is mainly a treeless landscape, where vegetation hugs the ground to survive freezing winds that contribute to blizzard conditions in winter, especially on alpine tundra. Elsewhere, the constantly frozen underlying ground

layer called permafrost (see p.175) blocks the movement of water through the soil during the summer thaw, turning some low-lying Arctic tundra areas into bogs. Despite these challenges, almost 2,000 plant species grow in the tundra, primarily grasses, sedges, mosses, and shrubs. All are shallow-rooted to cope with the tundra's thin active soil layer and to survive the region's freeze-thaw cycle.

Flowering plants such as saxifrage bloom vigorously during the brief summer, while mosses and lichens provide vital nourishment in lean months for mammals such as reindeer and Arctic hares. In areas where the soil is slightly deeper and richer, mat-forming shrubs such as bilberry and crowberry produce fruit in summer that may last beneath winter snows until the following spring.

## ▽ SUMMER COAT

The Arctic fox is well-adapted to its environment. In summer, its white winter coat thins, changing to brownish-grey to match the tundra's rocks and low-growing vegetation.

